# PRODUCT USER MANUAL

## For
## OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_bottom_salinity_arkona_bornholm

**Issue: 4.0**

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**Approval Date :** November 2025

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# RECORD TABLE

| Issue | Date | § | Description of Change | Author | Validated By |
|---|---|---|---|---|---|
| 1.0 | 2018/09/06 | all | Creation of the document | U. Raudsepp<br>I. Maljutenko<br>J. Linders | P. Lagemaa |
| 1.1 | 2019/02/21 | | Revision | U. Raudsepp<br>I. Maljutenko | P. Lagemaa |
| 2.0 | 2019/10/10 | | Update to 2018 data | U. Raudsepp<br>I. Maljutenko | P. Lagemaa |
| 2.1 | 2021/10/01 | all | Minor changes | P. Lagemaa | V. Huess |
| 3.0 | 2023/09/01 | all | Template change+ minor changes | P. Lagemaa | V. Huess |
| 4.0 | 2025/05/22 | all | Template change+ minor changes | P. Lagemaa | V. Huess |
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# GLOSSARY AND ABBREVIATIONS

| | |
|---|---|
| **MOI** | Mercator Ocean International |
| **NetCDF** | Network Common Data Form |
| **OMI** | Ocean Monitoring Indicator |
| **PUM** | Product User Manual |
| **QUID** | Quality Information Document |
| **MBI** | Major Baltic Inflow |

# DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

# 1) INTRODUCTION

## a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_bottom_salinity_arkona_bornholm. It contains one dataset:

- omi_health_tempsaloxy_baltic_mbi_bottom_salinity_arkona_bornholm

The OMI provides set of parameters which describe Major Baltic Inflows (MBI) in the southern and central Baltic Sea. The bottom salinity at the southernmost basins – Arkona and Bornholm Basins – shows first signals of inflow occurrence. In more general terms, time series of the bottom salinity in the Arkona Basin describes water exchange events between the Baltic Sea and the North Sea through the Danish Sounds. The 7-day moving average filtered bottom salinity is found based on daily mean Baltic Sea reanalysis product BALTICSEA_MULTIYEAR_PHY_003_011.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

# 2) DESCRIPTION OF THE PRODUCT SPECIFICATION

## a) General Information

| | |
|---|---|
| **Product Lines** | OMI_HEALTH_TEMPSALOXY_BALTIC_mbi_bottom_salinity_arkona_bornholm |
| **Geographical coverage** | Baltic Sea |
| **Variables** | Bottom salinity |
| **Available time series** | From 1993 to M-4 |
| **Temporal resolution** | Daily |
| **Horizontal resolution** | Two stations |
| **Integration depth (for example)** | Sea bottom |
| **Format** | NetCDF4 |

## b) Production System Description

The current Major Baltic Sea OMI consist of 7-day moving average filtered bottom salinity time series at southern Baltic basins. In principle those are the key parameters that show the signal of inflowing water mass to the Baltic Sea regardless of data source.

The bottom salinity dataset has been extracted from Copernicus Marine Service reanalysis for the Baltic [BALTICSEA_MULTIYEAR_PHY_003_011]. The location coincides with the main HELCOM monitoring stations BY1 and BY5 for Arkona and Bornholm Basins, respectively.

## c) Dataset details

| |
|---|
| **omi_health_tempsaloxy_baltic_mbi_bottom_salinity_arkona_bornholm** contains the following variables: |
| **sob_ark** [0.001]<br>long_name: Bottom Salinity at Arkona Basin<br>standard_name: sea_water_salinity_at_sea_floor<br><br>**sob_bor** [0.001]<br>long_name: Bottom Salinity at Bornhlom Basin<br>standard_name: sea_water_salinity_at_sea_floor<br><br>**time** [days since 1950-01-01 00:00:00]<br>long_name: time<br>standard_name: time |

# 3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

# 4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

# 5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-HEALTH-TEMPSALOXY-mbi-bottom-salinity-arkona-bornholm)